

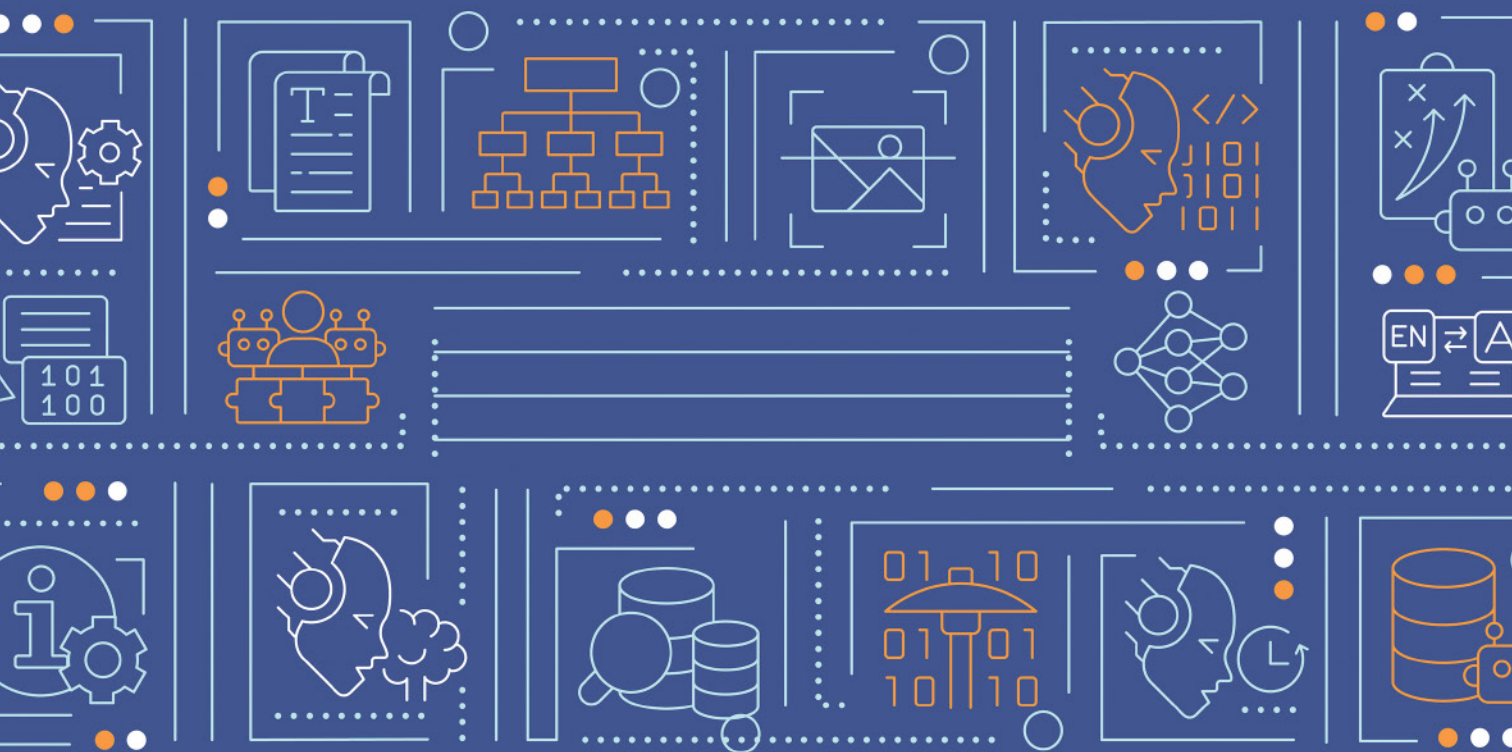


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LARGE LANGUAGE MODELS (LLMs) serve as critical foundations for a wide range of generative AI applications, which demand substantial memory and storage. The advent of monolithic 3D stackable DRAM, offering cost-effective vertical scaling akin to 3D NAND, presents a promising solution to this challenge. This article conducts a comparative analysis of a variety of 3D DRAM technologies with capacitor or capacitor-less storage mechanisms, focusing on vertical bitline (VBL) and vertical wordline (VWL) configurations. Key performance indicators, including access energy, latency, and storage density, are evaluated at the bank-level considering the 3D array core and peripheral circuits under

Monolithic 3D Stackable DRAM

An Enabling Technology for Large Language Model Acceleration

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Inference involves feed-forward operations in LLM comprising 2 steps: 'prefill' and 'decode'. Typically, when inference is performed, first feedforward is performed on the full input prompt and the keyvalue pairs are stored. This is known as 'prefill' operation and is performed only once for a new input.

the array. Moreover, a 3D DRAM-centric LLM accelerator is proposed and benchmarked against the off-the-shelf GPU platforms with high-bandwidth-memory (HBM) with die stacking of 2D DRAM

dies. The benchmark results underscore the potential of 3D DRAM technologies in enhancing the efficiency and scalability of LLMs, heralding a new era of large-scale AI model training and inference.

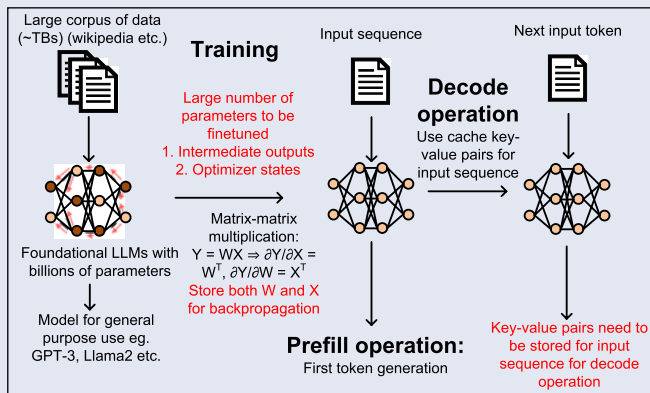


FIGURE 1 Large memory requirement for training and inference of LLMs. Training requires storing intermediate outputs and optimizer states such as momentum and variance. Inference comprise prefill and decode operation. In this work, we consider the effect of the training and decode operation. For decode, the key-value pairs are stored as shown later in Figure 2.

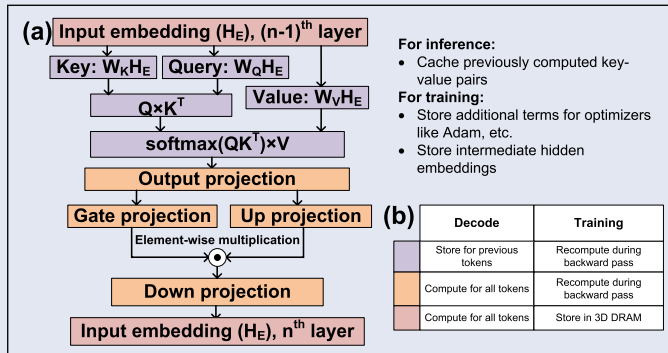


FIGURE 2 (a) Compute intensive layers in Llama-2 model [9] in a single layer, with a total of 32 layers (b) Different requirements for training and decode operation in terms of memory access and storage capacity.

I. INTRODUCTION

LLMs[1] are ubiquitous for generative language tasks. However, the model size for such foundational models is increasing rapidly and it is difficult to store the complete model on a single GPU's DRAM and require multiple GPUs. Apple researchers [2] proposed use of NAND Flash storage for LLM inference, which has slow read and write speed, thus not suitable for training. Training LLM models require large storage capacity (~5× model size) to store the intermediate data required for the backpropagation in training as in Figure 1. Recently, high-density monolithically 3D stackable DRAM [3], [4], [5], [6], [7], [8] has been proposed to continue the scaling of DRAM roadmap. In this article, 3D DRAM is referred to as monolithically 3D stackable DRAM with bit-cost-scalable fabrication approach. However, the 3D physical design and the comparative analysis is lacking in the literature. In this work, we evaluate emerging 3D DRAM technologies (capacitor-based and capacitorless-based) for potential 3D physical design and benchmark the bank-level performance. On top of the 3D DRAM technologies, we build 3D stacked systolic array-based accelerators for LLM including inference and pre-training. Finally, we present the LLM performance comparison across the 3D DRAM technologies and demonstrate ~15× throughput and ~55× energy efficiency compared to NVIDIA H100 GPU. The rest of the article is divided as follows. Section II describes the background of the LLM and its different operations for both training and inference. Section III describes the details about the monolithic 3D stackable DRAM technologies. We further discuss the detailed physical design of the 3D array and integrate it with the peripherals. The complete bank-level 3D DRAM is integrated with the logic die, described in Section IV. Section V describes the hardware evaluation and compares the performance for different design choices. Finally, we present a conclusion in Section VI.

II. BACKGROUND: LLM MODEL AND OPERATIONS

We chose Meta's open-source Llama2-7B [9] (7 billion parameter models) for

evaluations to study both training and inference. LLMs have different requirements for training and inference. Each of the operations have different trade-offs between memory access energy, logic recompute energy and data movement energy. We will describe the trade-off between various algorithms as follows:

A. INFERENCE

Inference involves feed-forward operations in LLM comprising 2 steps: ‘prefill’ and ‘decode’. Typically, when inference is performed, first feedforward is performed on the full input prompt and the key-value pairs are stored. This is known as ‘prefill’ operation and is performed only once for a new input. Next, the stored key-value pairs, also known as KV-cache, are utilized to generate subsequent output tokens. This operation is known as the ‘decode’ operation and requires storing a large KV-cache with a size proportional to the prompt length. The key-value pairs are different for each input; hence the size of the KV-cache increases linearly with batch size. Typically, batching is beneficial as it requires loading the models only once from the memory and provides better core utilization, but it also requires larger memory capacity for storing KV-cache. As a result, there is a trade-off between memory access energy and memory capacity available. Typically, models can be reduced to 8-bit integer type multiply-and-accumulate (MAC) operations, without significant reduction in the accuracy [10], which can greatly improve the performance.

B. PRE-TRAINING

Pre-training is similar to conventional backpropagation in neural networks. For pre-training, we need to store a large amount of data due to storing intermediate weight update matrices required for optimizers such as Adam, where an additional $2\times$ size of model parameters is required to be stored for weight updates. Further, to reduce training overheads, mixed precision training [11] is commonly used where the FP16 datatype is used for MAC operations but the weight update, momentum and variance are stored in 32-bit floating point (FP32). As a result, we need to store nearly $3\times$ model size in 32-bit precision. As a

The high voltage decoders for wordline are fabricated under the memory array that can tolerate high thermal budget for array formation. The low voltage circuits such as low voltage decoders, data buffers and control logics are fabricated in an advanced CMOS die and bonded with the 3D DRAM array. Given the limited bonding pads with pitch of $\sim 1\mu\text{m}$ for a mature hybrid bonding process, the 8:1 MUX can reduce the cross-tier contacts thus fit the bonding pads into the array area.

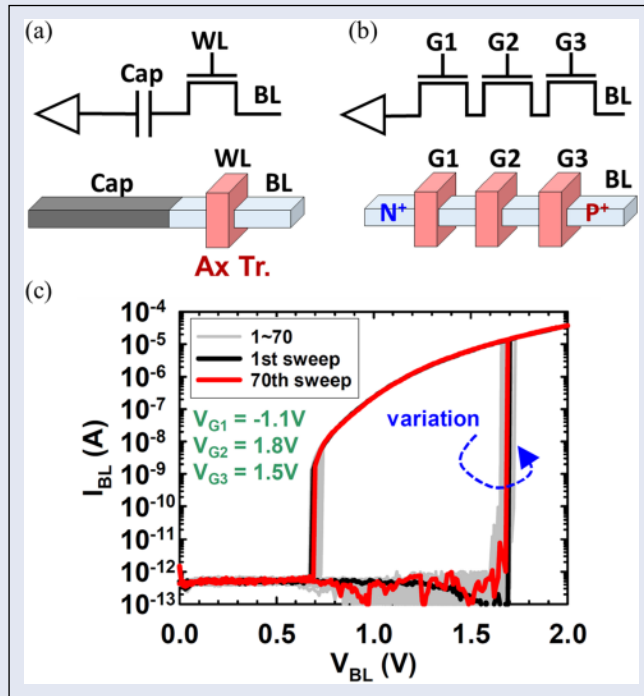


FIGURE 3 (a) Horizontal 1T1C schematic and device illustration. (b) Gate-control thyristor (GCT) schematic and device illustration. (c) The measured hysteresis I_{BL} - V_{BL} curves of a single GCT device [5]. The device can be programmed with bitline voltages.

result, the memory capacity required is typically very high for pre-training and training multiple inputs in parallel is not possible.

In this work, we consider inference and pre-training LLMs. For fair compari-

son between different technology nodes, we consider Llama2-7B for inference and training performance evaluations. We vary the batch size based on the amount of available storage to obtain the trade-offs between routing energy (data movement),

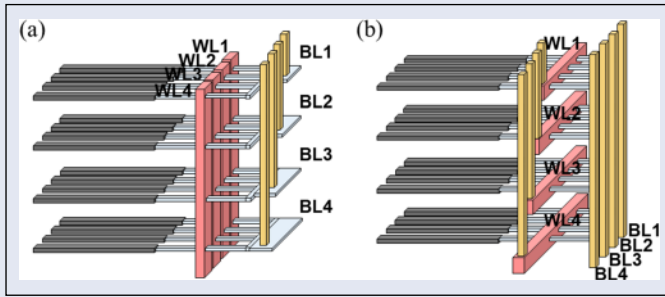


FIGURE 4 (a) Stacked 3D DRAM device with vertical wordline and horizontal bitline. (b) Stacked 3D DRAM device with vertical bitline and horizontal wordline.

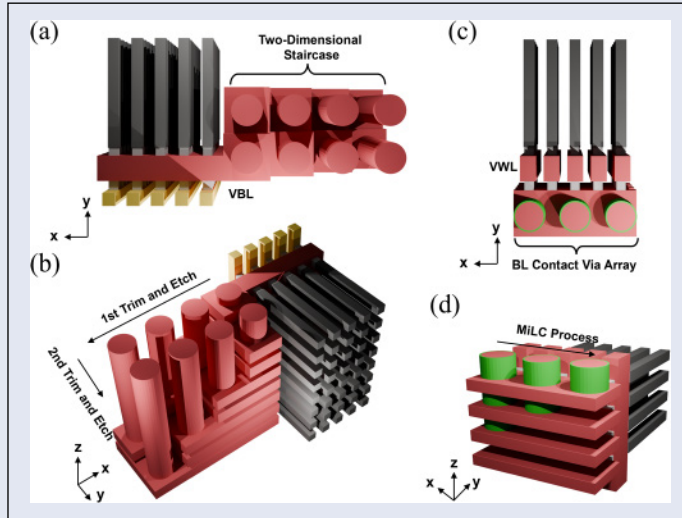


FIGURE 5 (a) Top view of the WL staircase for VBL routing design. The staircase can be fabricated in two dimensions. (b) Bird-eye's view of the 2D staircases fabricated with two directional trim and etch process. (c) Top view of the BL contact via array for VWL routing design. (d) Bird-eye's view of the BL contact via array fabricated with MiLC process for cost saving.

TABLE 1 Summary of horizontal 1T1C device dimension.

	H1T1C VWL [6]	H1T1C VBL [6]
Device X Pitch (nm)	80	70
Device Y Pitch (nm)	~700	~700
Device Z Pitch (nm)	70	100
Staircase Pitch (nm)	N/A	500
Contact Via pitch (nm)	300	N/A

layer re-computation (logic) and 3D DRAM access energy (memory access). It is observed that due to lower logic energy consumption in decode phase, higher 3D DRAM access limits the overall performance, whereas for training, the energy consumption is dominated by the FP16 MAC units. Hence, 3D DRAM's higher

capacity helps in performing computation for larger batch size.

III. MONOLITHIC 3D STACKABLE DRAM TECHNOLOGIES

As the scaling of 2D DRAM faces barriers in sub-10nm nodes, various 3D DRAM technologies have been pro-

posed [3], [4], [5], [6], [7], [8]. In this section, we reviewed the proposed 3D stackable DRAM candidates. On top of the 3D DRAM technologies, we design the physical placement for peripheral circuits and 3D interconnect routing to have a comprehensive technological study of monolithic 3D DRAM technologies.

A. 3D STACKABLE DRAM DEVICES

Although there are many proposed candidates for 3D DRAM, here we review the two most viable devices from industry. The proposed 3D stackable devices are

1) HORIZONTAL 1T1C (H1T1C)

The straightforward way is rotate the conventional 1T1C DRAM and stack on top of each other [3]. The process leverages the stacked nanosheet and 3D bit-cost-scalable processes [7]. The device schematic is shown in Figure 3(a).

2) GATE-CONTROL THYRISTOR (GCT)

Gate-control Thyristors leverage the band diagram engineering to realize memory characteristics. These device switch based on the avalanche mechanism that either provides super-steep sub-threshold switching for transistor [12] or long retention as a capacitorless DRAM [4], [5]. The device schematic and electrical characteristic is shown in Figure 3(b) and (c)[5].

B. 3D PHYSICAL ROUTING DESIGN

In order to realize a memory array, routability of the 3D stacked devices is crucial. Considering the fabrication, the physical designs for 3D DRAM routings are

1) VERTICAL WORDLINE (VWL) OR VERTICAL BITLINE (VBL)

The bitline and the wordline of the array can be placed vertically or horizontally. Therefore, there are two types of routing: **Vertical Wordline (VWL)** with horizontal bitline and **Vertical Bitline (VBL)** with horizontal wordline as shown in Figure 4.

2) STAIRCASES AND CONTACT VIAS

The routing of the vertically wired bitline and wordline is connected directly to the back-end-of-line (BEOL) metal

layers on top. However, the horizontally wired bitline and wordline requires pickup contacts toward BEOL metal layers. Staircases utilized in 3D NAND technology are suitable for wordline connectivity as shown in Figure 5(a). The trim and etch process is matured and can be done in two directions [13] as shown in Figure 5(b) to achieve compact staircase area and save lithography cost. On the other hand, bitline pickup contact could also be fabricated in the staircase fashion. Nevertheless, the large bitline area with large capacitance affects the sensing margin. Alternatively, a compact structure with BL contact vias as shown in Figure 5(c) can reduce the pitch and area [6]. As shown in Figure 5(d) BL contact vias vertically etched through the layers of BL and land on the target BL layer and the contact is wrapped with a sidewall isolation for electrical separation between contact vias and the bitline layers. Note that the cost of the lithography steps is increased proportional to the number of layers, which can be alleviated by MiLC process [14].

C. HORIZONTAL 1T1C (H1T1C) 3D DRAM

Combining the 3D stackable device with the 3D physical routing designs, the 3D DRAM memory array for horizontal 1T1C can be designed into two configurations:

1) HORIZONTAL 1T1C – VERTICAL WORDLINE (VWL) [3], [6]

Figure 6 illustrates the H1T1C VWL memory array. The wordline is vertically connected and the bitline contact via array is employed. According to the prior device study [6], the gate length for the access transistor to be 100 nm and the length of the capacitor is 500 nm to have enough sensing margin. The dimensions are concluded in Table 1.

2) HORIZONTAL 1T1C – VERTICAL BITLINE (VBL) [3], [6], [8]

Figure 7 illustrates the H1T1C VBL memory array. The bitline is vertically connected. Similar to H1T1C VWL, the device dimensions are summarized in Table 1. Note that the wordline pickup contact is staircase design with the pitch of 500 nm realized in mature 3D NAND technologies. As shown in

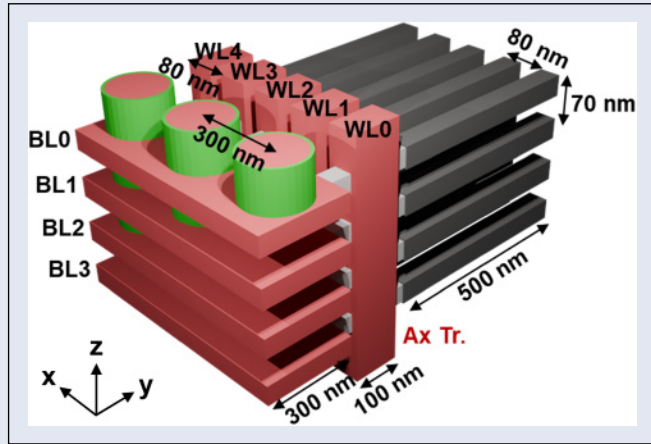


FIGURE 6 Horizontal 1T1C with vertical wordline. From the reported physical dimensions [6], the BL pitch is 80 nm to have enough separations. The gate length is 100 nm for access transistor and the length of cap is 500 nm for enough sensing margin. The pitch of the BL contact via is 300 nm.

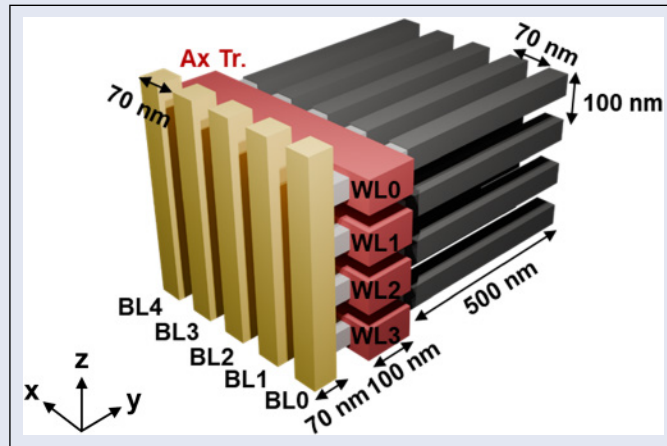


FIGURE 7 Horizontal 1T1C with vertical bitline. From the reported physical dimensions [6], the dense BL pitch is 70 nm. The gate length is 100 nm for access transistor and the length of capacitor is 500 nm for enough sensing margin.

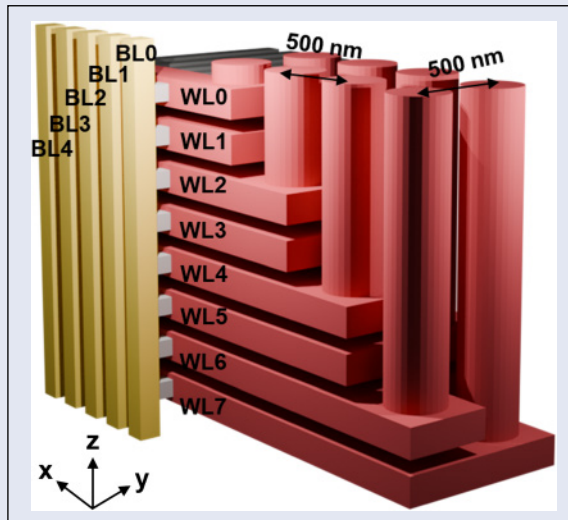


FIGURE 8 Two-dimensional wordline staircase attached to the horizontally wired wordline. The pitch of the staircase is 500 nm.

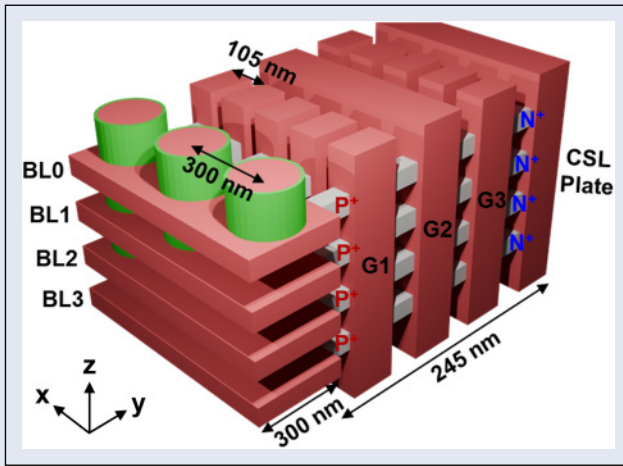


FIGURE 9 Gate-controlled thyristor with vertical wordline. The reported BL pitch [4] is 105 nm for vertical gate separation. The total length of the device is 245 nm and the BL contact via pitch is 300 nm.

TABLE 2 Summary of gate-controlled thyristor device dimension.

	GCT VWL [4]	GCT VBL [5]
Device X Pitch (nm)	105	52.5
Device Y Pitch (nm)	245	245
Device Z Pitch (nm)	70	70
Contact Via pitch (nm)	300	300

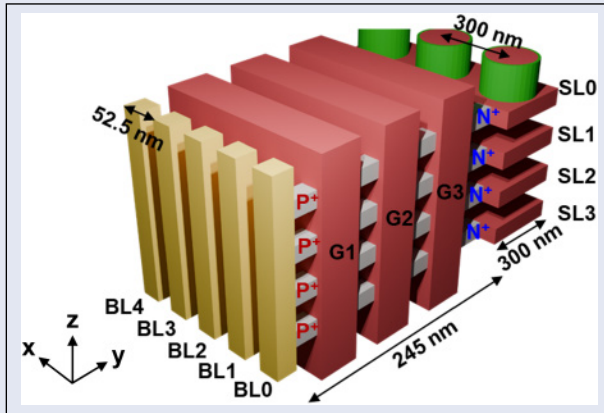


FIGURE 10 Gate-controlled thyristor with vertical bitline. The reported BL pitch [5] is 52.5 nm denser than the vertical wordline design. The total length of the device is still 245 nm, and the SL contact via pitch is 300 nm.

Figure 8, the wordline staircase is a two-dimensional staircase array.

D. GATE-CONTROLLED THYRISTOR (GCT) 3D DRAM

On the other hand, GCT device can also be classified into two fashions by routing:

1) GCT – VERTICAL WORDLINE (VWL) [4]

Figure 9 illustrate the GCT VWL memory array. The wordline of G1/G2/G3 is vertically connected and the bitline contact via array is employed. Unlike HITIC, GCT VWL is a capacitor-less design and we propose to leverage the

bitline contact via array for bitline pickup contact. The dimensions are summarized in Table 2.

2) GCT – VERTICAL BITLINE (VBL) [5]

Figure 10 illustrates the GCT VBL memory array. Unlike the HITIC VBL memory array shared with the common ground plate, GCT VBL utilizes the sourceline to form a 3D crossbar structure for cell selection. To have a compact cell area, we also propose the source contact via array. The bitline is vertically connected. Similar to HITIC VWL, the device dimensions are summarized in Table 2.

E. 3D ARRAY CONFIGURATION

To evaluate the 3D DRAM technologies, we design the bank-level DRAM array for four of the devices. The 2D schematic is shown in Figure 11. The 3D DRAM bank contains 32 subarrays of which consist of 32 MATs. We define a single MAT with common DRAM design of 1024 WL and 1024 BL. In the MAT level, we design the local WL switch matrixes to select local row with large RC parasitic. Moreover, 8:1 MUXes are deployed after the sensing circuits for selectively read out the column data. Since the 3D stackable DRAM leverages the 3D NAND fabrication techniques, we employ two mature 3D integration schemes, CMOS under array (CUA) [15] and Cu-Cu hybrid bonding CMOS directly bonded to Array (CBA) [16] for a compact form factor. Figure 12 illustrates the general 3D DRAM array with 3D integrations. The high voltage decoders for wordline are fabricated under the memory array that can tolerate high thermal budget for array formation. The low voltage circuits such as low voltage decoders, data buffers and control logics are fabricated in an advanced CMOS die and bonded with the 3D DRAM array. Given the limited bonding pads with pitch of $\sim 1\mu\text{m}$ [17] for a mature hybrid bonding process, the 8:1 MUX can reduce the cross-tier contacts thus fit the bonding pads into the array area. Note that the sensing circuits placement for HITIC and GCT is different here. With respect to the high sensitivity of sensing margin

for 1T1C DRAM, the sensing circuits are deployed on the memory tier to avoid extra parasitic bitline capacitance through Cu-Cu bonding pads. On the other hand, the GCT device reads the bitline current and would not suffer from the parasitics of bonding pads. The sensing circuits can thus be deployed after the MUXes and on the advanced CMOS tier for better performance.

Figure 11 also illustrates the simulation framework for bank-level 3D DRAM array performance. The 3D DRAM array physical parameters are collected from Coventor [6], [16] process model and the peripheral circuits are extracted from NeuroSim [19] in 32 nm technology node for CUA and 7nm technology node for CBA. The bank-level performance for various 3D DRAM technologies is summarized in Table 3. We compare 64 layers and 128 layers for the LLM accelerator designs. Among HIT1C, VWL with 128 layers achieves the smallest bank area. Although VWL is better than VBL when scaled to 128 layers, the sensing margin of the VWL would be significantly decreased due to the large horizontal BL capacitance. For GCT, VBL design has the smallest form factor for 128 layers and the lowest read energy due to the low voltage operation [5]. GCT VBL with 128 layers is the best candidate in terms of memory bit density (~ 2.74 Gb/mm²).

IV. 3D DRAM-CENTRIC ACCELERATOR FOR LLM

Next, we design a 3D stacked LLM accelerator, comprising a logic base die integrated with the 3D DRAM.

A. SYSTOLIC ARRAY

A digital systolic array logic base die is designed for MAC computations in model by the GAMMA [20] and data access pattern is generated by the MAESTRO [21] tool. The power, performance and area (PPA) for the logic base die is estimated by circuit-level simulations from TSMC 40 nm PDK and scaled to advanced node such as 7 nm and 3 nm using the NeuroSim [19] tool. A single tile is designed for each layer in the forward and backward pass and it is pipelined and duplicated for maximum throughput. We consider a separate design of systolic array inference and inference

A single tile is designed for each layer in the forward and backward pass and it is pipelined and duplicated for maximum throughput. We consider a separate design of systolic array inference and inference to find the major bottlenecks in each of the case. Finally, heterogeneous integration is employed to stack the 3D DRAM on top of the logic base die with TSVs.

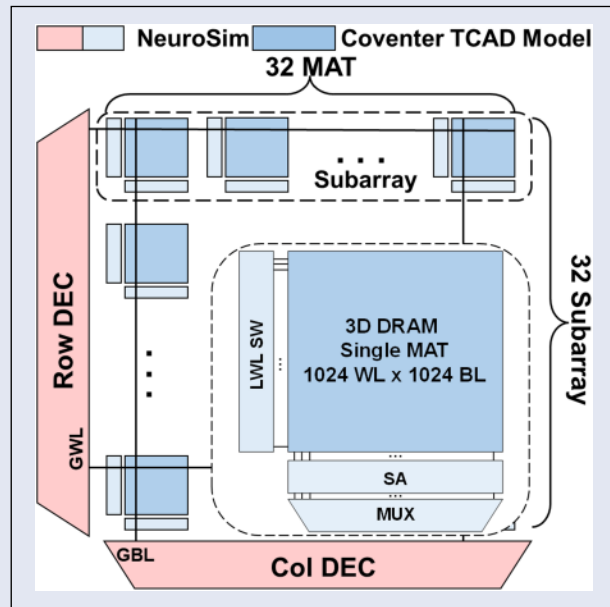


FIGURE 11 2D bank-level schematics of 3D DRAM array. A 3D DRAM bank contains 32 by 32 MATs. A single MAT has 1024 WL and 1024 BL. To avoid large WL RC parasitic from 3D design, local wordline switch matrixes are deployed in MAT level. The MUX is designed are sensing circuits to select the column for read out. Performance is calculated with NeuroSim [19] and Coventor process simulator [18].

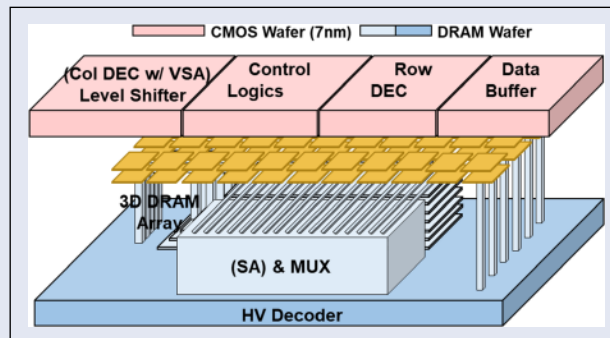


FIGURE 12 3D DRAM array with heterogeneous integration, hybrid-bonding and CMOS-under-array, for a compact form factor. The sensing circuit placement for HIT1C and GCT is differentiated in the consideration of sensing performance. For HIT1C, sense amplifier (SA) is placed on the same tier. For GCT, the sensing circuit is placed on the top advanced CMOS tier for better performance.

TABLE 3 Bank-Level performance comparison for various 3D DRAM technologies.

BANK-LEVEL PERFORMANCE	H1T1C VWL		H1T1C VBL		GCT VWL		GCT VBL	
Layers	64	128	64	128	64	128	64	128
MAT size	1024 WL x 1024 BL							
No. of MATs	32 x 32							
Capacity (Gb)	1							
Area (mm ²)	1.44	0.77	1.34	0.95	1.26	0.73	0.63	0.37
Memory Density (Gb/mm ²)	0.69	1.3	0.75	1.05	0.79	13.7	1.59	2.74
Read /Write Latency (ns)	~34	~34	~30	~31	~15	~15	~10	~10
Read Energy (pJ/bit)	3.18	3.37	3.26	3.54	6.26	7.18	2.76	3.05
Write Energy (pJ/bit)	3.18	3.37	3.26	3.54	9.22	10.9	3.5	3.79

to find the major bottlenecks in each of the case. Finally, heterogeneous integration is employed to stack the 3D DRAM on top of the logic base die with TSVs. The peripherals of 3D DRAM are also stacked through heterogeneous integration under and above the memory array as shown in Figure 13. Further, we limit the total power consumption below 200 W for the complete system to reduce the cooling overheads. Due to this, it is observed that even with higher storage capacity, the 3D DRAM options with higher access energy might lead to a lower overall throughput.

B. HETEROGENOUS INTEGRATION

3D heterogenous integration is performed between the logic base die and 3D DRAM by using TSV and Cu-Cu hybrid bonding, as described in [22]. The required TSV pitch on the low voltage peripheral dies is 10 μm to obtain a bandwidth density of 10000 Gbps/mm². The memory and logic dies are designed to have balanced area for each tier. Table 4 provides details about both the memory and logic tier.

Memory available can be scaled by the area of the memory tier.

V. HARDWARE PERFORMANCE EVALUATION

Key simulation parameters are also summarized in Tables 4 and 5. Here, we use the metric output tokens/s for performance and output tokens/J for energy efficiency. We cluster of such accelerators to store the model parameters and the intermediate data as required by the algorithm. Here, we chose a small model such that it can fit on the baseline ‘1b’ node DRAM as well for a fair comparison.

For inference, different batch sizes are considered based on the amount of storage available to store the KV-cache on 3D DRAM. For a fair comparison, we consider the baseline as a similar logic die 3D stacked with today's 2D DRAM die in 1b node (0.455 Gb/mm²) [23]. Figure 14(a) shows the total power savings of various 3D DRAM technologies compared to the baseline. A larger batch size can process more tokens in parallel as it reduces the weight reloading. As a result, we see an improvement in performance and energy efficiency for GCT 3D

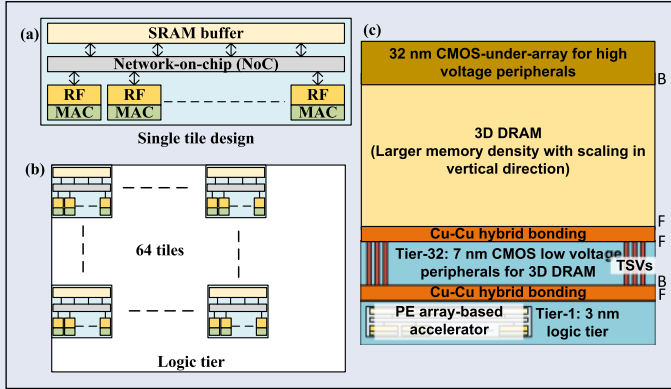


FIGURE 13 (a) A single tile design comprising, MAC units, register files (RF), a network on chip (NoC) and an SRAM. (b) The tile is repeated and pipelined for maximum throughput to obtain the complete chip (c) 3D stacked accelerator design with the 3D DRAM memory stacked on top of logic die.

TABLE 4 Simulation parameters for LLM accelerator.

300 mm ² DIE AREA	PARAMETERS	H1T1C VWL	H1T1C VBL	GCT VWL	GCT VBL
Logic Tier	Technology	3 nm FinFET Process			
Low V Peri. Tier	Technology	7 nm FinFET Process			
	Feature Size	35 nm			
	BL Pitch (nm)	80	70	105	52.5
	WL Pitch (nm)	700	700	245	245
	Staircase pitch	500 nm (two dimensional)			
3D DRAM	No. of BL/WL	1024/1024 (in a MAT)			
Physical	No. of MAT	32 x 32 (in a bank)			
Parameter	No. of Layers	64/128			
	Bank Size	1 Gb			
	Capacity (Gb)	64L	208	224	238
		128L	391	315	410
3D DRAM	WL Read Voltage (V)	3	3	-2/3/3	-1.1/1.8/1.5
Physical					
Parameter	BL Read Voltage (V)	1.2	1.2	2.5	1.5
CUA Tier	Technology	32 nm Process			

DRAM as it can store larger KV-cache and hence can run several samples in parallel, increase

For the HITIC, reduced performance benefits are observed due to higher access energy of memory. We observe that the performance and energy efficiency degrade for the decode case due to higher read energy for HITIC DRAM, even with a larger batch size.

For training, since we cannot fit all the model weights and optimizer states on single accelerator, we consider 12 such accelerators. Further, we consider different batch sizes like inference, however, here the DRAM energy consumption is not a bottleneck due to higher power consumption of FP16 logic core. Since each input requires storing additional intermediate layer outputs for backpropagation, the amount of storage required grows linearly with the batch size. We observe the trade-off between larger memory capacity and high energy consumption of the 3D DRAM memory. Figure 14(b) shows the comparison for different 3D DRAM memories.

Due to peak power limit of 200 W, the performance is limited by the peak power consumption. We observe higher performance improvements for the training with 3D DRAM as the logic core unit bottlenecks the overall power consumption, hence with a higher batch size we obtain a more efficient logic core design and a higher core utilization. However, for the case of inference, the performance is limited by the DRAM access energy, which is higher in the case of 3D DRAM. Hence, the overall performance gains are not as high.

Finally, we compare our proposed 3D DRAM-centric LLM accelerator with NVIDIA H100 GPU [24] for inference in Table 6. 3D DRAM-centric LLM accelerator achieves 54k tokens/s throughput and 289 tokens/J energy efficiency for the inference case. Compared to NVIDIA H100, with similar memory capacity, 3D DRAM-centric LLM accelerator achieves ~15 $\times$ throughput and ~55 $\times$ energy efficiency.

VI. CONCLUSION

This article surveyed the proposed 3D DRAM technologies and addressed

TABLE 5 Systolic array-based accelerator architecture specifications.

	INFERENCE	TRAINING
No. of MAC (No. of MAC/PE $\times$ No. of PE/core $\times$ No. of core/accelerator $\times$ No. of accelerators)	$32 \times 8192 \times$ $64 \times 1 = 16.8\text{M}$	$32 \times 4096 \times$ $16 \times 12 = 25.2\text{M}$
L1 size (kB) per PE	1	1
L2 size (MB) per core	1	10
Bit precision for MAC operation	8-bit	16-bit
TSV pitch (μm)		10
Cu-Cu hybrid bonding pitch (μm)		1

TABLE 6 Comparison between the designed 3D DRAM-centric accelerator and NVIDIA H100 accelerator.

	3D DRAM-CENTRIC ACCELERATOR	NVIDIA H100 GPU
Technology node (nm)	3 nm	TSMC's 4N node
Logic die area (mm^2)	250	814
DRAM Technology	GCT-VBL 128 layers	5 stacks of HBM3
Total memory (GB)	84 GB	80 GB
Throughput (tokens/s)	54331	3626
Energy efficiency (tokens/J)	289	5.18
Datatype	INT8	FP8

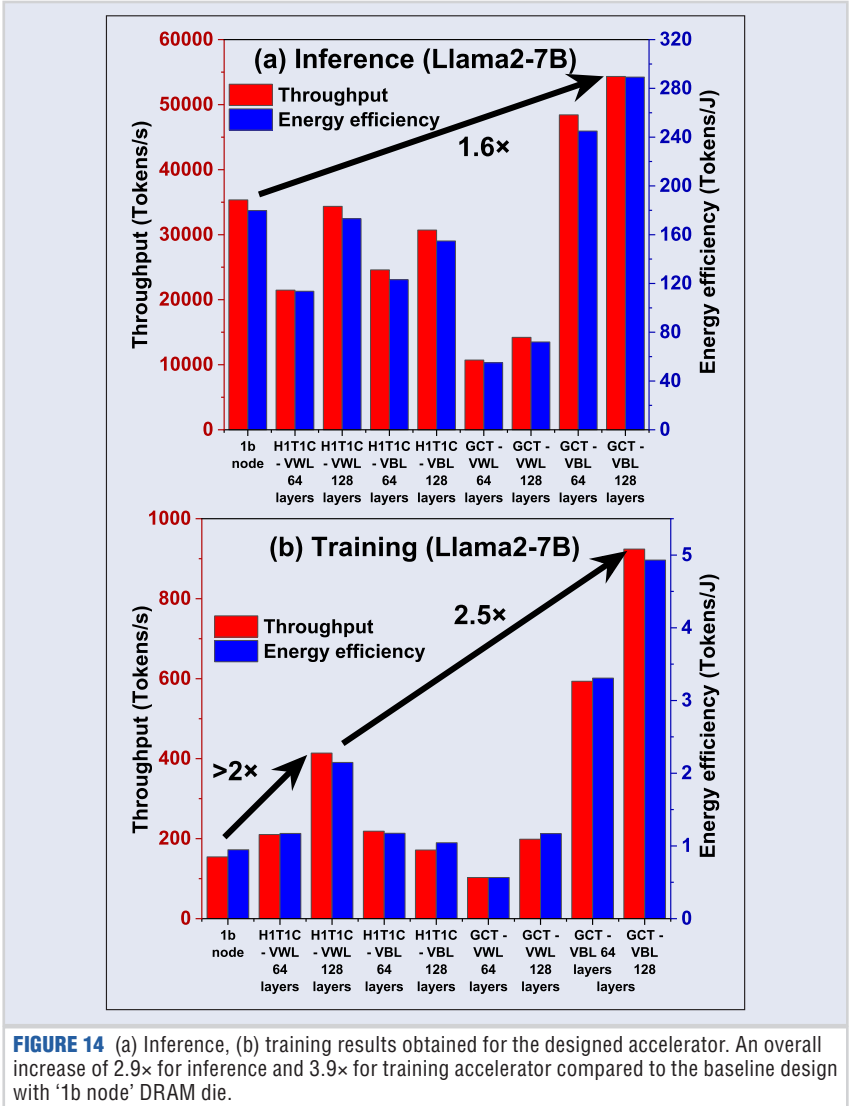


FIGURE 14 (a) Inference, (b) training results obtained for the designed accelerator. An overall increase of 2.9 $\times$ for inference and 3.9 $\times$ for training accelerator compared to the baseline design with '1b node' DRAM die.

the open questions for 3D physical designs. We also designed the 3D DRAM memory bank and provided bank-level performance. Based on the 3D DRAM memory bank design, we proposed three types of LLM accelerators for inference and training. Overall, our proposed 3D DRAM-centric LLM accelerator achieves 54k tokens/s throughput and 289 tokens/J energy efficiency for the inference case. Compared to NVIDIA H100, 3D DRAM-centric LLM accelerator achieves $\sim 15\times$ throughput and $\sim 55\times$ energy efficiency.

Although this article presents array designs based on 3D DRAM technologies. There are several challenges that remain to be addressed. First, although GCT outperforms H1T1C, there is potential device aging due to the latching process. Further extensive reliability verification is needed. Second, an alternative fabrication method [8] for 3D DRAM is proposed very recently. The mainstream fabrication of the future 3D DRAM structures could be different, leading to redesign of the 3D DRAM bank. Lastly, further vertical stacking would increase the staircase overhead and reduce the sensing margin. For scaling beyond 200 layers, the array needs to be redesigned for better area efficiency and tradeoff with access speed. For potential larger array, tiered memory design [25] is needed for different RC latencies across different layers. With the tiered memory design, the memory bandwidth can be improved by optimized bank interleaving with different tiered delays. We hope this article sheds some lights for future 3D DRAM technology-system co-design.

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